

PHENOVA Competitive Edition — Complete Answer Sheet

1. Platform

PHENOVA will be developed as an Android-first platform, with the architecture kept ready for a future iOS version.

2. Product Scope

PHENOVA will be developed as the full Competitive Edition rather than a limited MVP. The complete architecture will be established from the beginning, while implementation will prioritize getting the core systems working and tested end-to-end before expanding into less critical areas.

3. Editing Engine

The editing architecture will use a hybrid approach. The professional editing experience will be presented inside the Android application while web-based editor technology may be used where it provides a stronger implementation path. Heavy video processing and rendering will be handled by backend infrastructure rather than depending entirely on the user's Android device.

4. Supported Media

PHENOVA will support all practical editable media formats, including common video formats such as MP4, MOV, WebM, MKV and AVI; audio formats such as MP3, WAV, AAC, M4A and FLAC; and image formats such as JPG, JPEG, PNG, WebP and GIF. Additional practical formats should be supported where technically possible, with compatible media normalized when necessary.

5. Backend Architecture

The backend will follow a free-first hybrid strategy. PHENOVA should use multiple providers where appropriate and select the best available free infrastructure for each responsibility. The architecture should avoid unnecessary dependence on a single provider and should allow individual services to be replaced without rebuilding the entire application.

6. Authentication

Authentication will use the best available free architecture while maintaining a custom PHENOVA account system. Users should be able to use PHENOVA's own username, email and password authentication, while Google Sign-In will also be supported. The architecture must account for production Android distribution through Google Play and appropriate Google authentication requirements. Phone verification should be available as an optional capability.

7. Database and Storage

The database and storage layer will use a free-first hybrid architecture. PHENOVA needs to store users, projects, project settings, timelines, transcripts, AI edit plans, usage information, subscriptions, rendering jobs and media metadata. Large media files should be separated from normal application data and handled through appropriate object/file storage.

8. UI Design

PHENOVA should have a distinctive original identity rather than simply copying another editor. It should incorporate strong usability concepts from professional applications such as Premiere and mobile editors such as CapCut, while maintaining an original PHENOVA visual language and creator-focused mobile experience.

9. Editor Workspace

The editing workspace should be a full professional workspace adapted for Android. It should include a timeline, synchronized transcript, video preview, media/assets area, effects and tools, audio controls, captions, overlays and other relevant editing surfaces. Timeline and transcript editing should remain synchronized.

10. Export

PHENOVA should support all major social-media aspect ratios and practical CapCut-style presets, while also supporting custom aspect ratios. Export quality should support 720p, 1080p, 2K and 4K, subject to source material and available rendering infrastructure.

11. AI Capabilities

The AI system will support speech-to-text, transcription, transcript editing and regeneration, automatic captions, scene and clip analysis, object/media analysis, AI-assisted video and audio editing, and the AI Director/NLE planning system. The AI Director should understand user descriptions, uploaded footage, project clips, timeline, transcript, audio, instructions and previous edits.

12. AI Execution

AI execution will use a hybrid architecture. Lightweight operations that are practical on Android can run locally, while computationally intensive operations will run through backend infrastructure. The division should be determined by performance, privacy, device capability and free-infrastructure constraints.

13. Plan System

PHENOVA's existing Basic, Pro and Premium plan system remains authoritative. The professional editor may expose the full editing architecture, but actual capabilities, usage limits, duration limits, correction limits, quality restrictions and other plan restrictions must be enforced by the backend. Limits should be remotely configurable so they can be changed without requiring a new APK.

14. Rendering

Server-side FFmpeg will be the primary rendering foundation. FFmpeg will handle media processing such as cuts, transformations, audio processing, captions, resizing and encoding. The rendering layer should remain modular so that a more advanced proprietary rendering system can be introduced later without redesigning the entire platform.

15. First End-to-End Deliverable

The first end-to-end implementation should cover the full core PHENOVA workflow: account creation and authentication, project creation, media import, media processing, AI analysis, AI Director planning, synchronized timeline and transcript creation, manual editing, AI-assisted corrections, preview generation, rendering and export, with the appropriate plan and usage restrictions enforced throughout the process.

Overall Direction

PHENOVA should establish the full Competitive Edition architecture from the beginning while intelligently prioritizing the components required to make the core workflow function. The goal is a functioning AI video-editing platform with a professional editing architecture, AI Director, synchronized project model, scalable free-first backend infrastructure, secure authentication, plan enforcement and an expandable rendering pipeline.